# The Selling closure: reindexing measured lattices with a completeness guarantee

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**Selling reduction; obtuse superbase; Delone sorts; lattice symmetry; reindexing; pseudo-symmetry; groupoid**

## Abstract

A monoclinic crystal measured twice can reduce once to a cell on $c$ and once to a cell on $a + c$. The integer matrix between the two cells is not a lattice symmetry, so a reindexing that searches only symmetries cannot propose it. This paper treats all Selling-reduced cells of a lattice, with the unimodular matrices between them, as one finite object, the Selling closure; all its matrices lie in a fixed set of 6096. For a measured cell the closure is widened to every reduced cell within a strain $\epsilon$, in a norm that is the same in every basis. The widened set is finite and is enumerated exactly. Reindexing then returns every proper unimodular operator whose price, the strain by which it fails to match the two cells, is at most $\epsilon$, grouped into the distinct indexings, and offers a pseudo-twofold at twice the distance to the metrics it would fix. In 780 constructed cases, 120 cases with errors near the band and 210 real cells, the true operator was always found. Small-matrix and cctbx searches found it nearly as often, so the gain is the guarantee and the structure of the candidate set.

## 1. Introduction

Take a primitive monoclinic lattice with $a = 30$, $b = 40$, $c = 50\,\text{Å}$ and $\beta = 100°$. Its Selling-reduced cell has two right angles, $\alpha$ and $\gamma$, between vectors of the reduced superbase, and a measured cell lies slightly to one side or the other of each. One data set therefore reduces to the cell 30, 40, 50 Å, $\beta = 100°$, and another, equally correctly, to 30, 40, 53.7 Å, $\beta = 113.4°$, built on $a$ and $a+c$ (Section 3.2). The matrix between the two is not a symmetry of the lattice. A pipeline that idealises each cell to its Bravais lattice (Le Page, 1982; Sauter *et al.*, 2004) and standardises it will usually return the same conventional cell from both; a program that matches the two reduced cells directly must find that matrix, and does so by trying a list of small integer matrices, or the matrices that unit-cell similarity searches generate (Grosse-Kunstleve *et al.*, 2002). Such a search succeeds when the list contains the operator. This paper replaces the list with a set that provably contains every operator within a stated tolerance.

The multiplicity has a classical description. A superbase is four lattice vectors summing to zero; it is obtuse, or Selling-reduced, when all six pairwise dot products are non-positive (Selling, 1874). A superbase is obtuse exactly when its partial sums are Voronoi vectors of the lattice (Voronoi, 1908; Conway & Sloane, 1992), and the obtuse superbase is unique except when a dot product vanishes (Delaunay, 1933; Patterson & Love, 1957). Kurlin (2022) listed all obtuse superbases of each Voronoi type and derived from them a complete isometry invariant. Delone's 24 sorts classify lattices by the zeros and equalities among the six dot products (Delone *et al.*, 1975; Burzlaff & Zimmermann, 2006; Andrews & Bernstein, 2023). The $S^6$ representation of Andrews *et al.* (2019*b*) writes the same six numbers as a point of a six-dimensional space; its boundary transformations and reflections are the operators between reduced presentations of a lattice, and SELLA (Andrews *et al.*, 2023) measures the distance from a cell, through those images, to the subspace of each sort, the projection approach that

goes back to Andrews & Bernstein (1988). Oishi-Tomiyasu (2012, 2024) gives error-stable Bravais-lattice determination in Selling coordinates, with a bound under which a finite list of basis transforms provably suffices. On the Niggli side, the multiplicity of Buerger cells (Buerger, 1960; Gruber, 1973) and the instability of the reduced cell under error (Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) are related phenomena. The operators that relate two settings of one lattice form a coset of its symmetries (Flack, 1987).

What this paper takes from that work is the operators and the sorts. What it adds is:

1. the explicit set $\mathcal{U}$ of 6096 integer matrices that contains every operator between two reduced cells of any lattice, its layers, and the closures of the 24 sorts computed from it, with the number of symmetries that each reduced cell shows and the number of cell shapes (Sections 3 and 5.1);

2. for a measured cell, the widened closure in a strain norm that is the same in every basis, shown to be finite and enumerated exactly (Section 4);

3. reindexing from one reduction of the new cell, with a guarantee that every operator within the band is among the candidates, the coset structure of the candidates and their classes under the Laue group, and the rule by which a pseudo-symmetry is priced and offered (Sections 4.5 and 5.3).

The practical gain is the guarantee, not the hit rate: on the cases tested, a search over small integer matrices between the two reduced cells finds the true operator almost as often (Section 5.3).

Section 2 collects the classical definitions. Section 3 defines the closure and works through the monoclinic cell. Section 4 gives the methods for measured cells and states what is proved. Section 5 gives the closures of the 24 sorts, the completeness of the

computation, the reindexing results and the distances of a measured cell to the sorts. Proofs are in Appendix A. The supporting information holds the catalogues of shapes and specialisations, the construction of the operator set with no lattice, the theory of an absolute band on the conorms, further tables, and a list of the computational checks keyed to the sections of this paper.

## 2. Background

### *2.1. Superbases, conorms and the metric tensor*

Selling reduction works with four vectors: the cell vectors $a$, $b$, $c$ and $d = -a - b - c$. The four form a *superbase*, and any three of them are a basis of the lattice. The six negated pair products

$$p_{da} = -d \cdot a, \quad p_{db} = -d \cdot b, \quad p_{dc} = -d \cdot c, \quad p_{ab} = -a \cdot b, \quad p_{ac} = -a \cdot c, \quad p_{bc} = -b \cdot c$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992). Conorm vectors are always written in this slot order, $da, db, dc, ab, ac, bc$; the Selling parameters of $S^6$ are the same numbers with the opposite sign, in another order. A superbase is *obtuse*, which is to say Selling-reduced, when all six conorms are non-negative. The metric tensor in the basis $(a, b, c)$ is linear in the conorms,

$$\mathbf{G} = \begin{pmatrix} p_{ab} + p_{ac} + p_{da} & -p_{ab} & -p_{ac} \\ -p_{ab} & p_{ab} + p_{bc} + p_{db} & -p_{bc} \\ -p_{ac} & -p_{bc} & p_{ac} + p_{bc} + p_{dc} \end{pmatrix}, \tag{1}$$

so the conorms are six cell parameters. A zero conorm is a right angle between two of the four vectors; the set of slots that vanish is the *zero pattern*, and the set of conorm vectors with $p \geq 0$ and those slots zero is a *face* of the cone of non-negative conorms. A face describes lattices only if its non-zero slots connect the four vectors. Up to relabelling there are six kinds of face: no zero, one zero, two zeros on disjoint pairs, two zeros sharing a vector, and three zeros forming a triangle or a path. There are 38 faces in all. The triangle and the path describe the same lattices, those with

three mutually perpendicular lattice vectors, read in different cells, so the six kinds of face are the five Voronoi types (Voronoi, 1908; Conway & Sloane, 1992). A *frame* is a choice of basis in which a metric tensor or a superbase is written.

### *2.2. Flips and swaps*

Two kinds of integer operator act on a superbase.

*Flips.* A flip at the conorm $p_{ab}$ replaces $a$ by $-a$ and adds the old $a$ to the two other vectors, $(a, b, c, d) \mapsto (-a,\ b,\ c + a,\ d + a)$. The new conorms are $p'_{ab} = -p_{ab}$, $p'_{ac} = p_{da} + p_{ab}$, $p'_{da} = p_{ac} + p_{ab}$, $p'_{bc} = p_{bc} + p_{ab}$, $p'_{db} = p_{db} + p_{ab}$ and $p'_{dc} = p_{dc} - p_{ab}$. Selling reduction applies flips at negative conorms until none remain. When $p_{ab}$ is exactly zero the flip exchanges the slots $ac$ and $da$ and carries one obtuse superbase to another of the same lattice, which is why the reduced cell of a lattice with a right angle is not unique. In the basis $(a, b, c)$ each flip is a fixed $3 \times 3$ integer matrix of determinant $-1$ that is its own inverse (Table 1). A word of flips such as *ab.bc.db* is applied left to right, each flip being read in the labelling current at that step, and its matrix is $\mathbf{F}_{ab}\mathbf{F}_{bc}\mathbf{F}_{db}$, columns being the new basis vectors in the old basis.

Table 1. *The six Selling flips as* $3 \times 3$ *integer matrices in the basis* $(a, b, c)$, $d = -a - b - c$; *columns are the new basis vectors in the old basis. Each has determinant* $-1$ *and squares to the identity. For the flips at* $da$, $db$ *and* $dc$ *the vector* $d$ *is negated and the new* $a, b, c$ *are listed.*

| flip at | new basis | matrix |
|---|---|---|
| $ab$ | $(-a,\ b,\ a+c)$ | $\begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ |
| $ac$ | $(-a,\ a+b,\ c)$ | $\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ |
| $bc$ | $(a+b,\ -b,\ c)$ | $\begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ |
| $da$ | $(a,\ -a-c,\ -a-b)$ | $\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}$ |
| $db$ | $(-b-c,\ b,\ -a-b)$ | $\begin{pmatrix} 0 & 0 & -1 \\ -1 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$ |
| $dc$ | $(-b-c,\ -a-c,\ c)$ | $\begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ -1 & -1 & 1 \end{pmatrix}$ |

*Swaps.* A superbase is an unordered set of four vectors up to a common sign, so one reduced cell can be written in $4! \times 2 = 48$ ways. The 48 reorderings, with or without a change of the common sign, are the swaps. They change which vectors are called $a, b, c$ and nothing else. Negating a single basis vector is not a swap: it does not preserve $a + b + c + d = 0$.

*In* $S^6$. A flip is a boundary transformation of $S^6$ (Andrews *et al.*, 2019*b*), written here on the basis rather than on the six parameters. A swap acts on the six Selling parameters as one of the 24 reflections of $S^6$; the common sign change acts trivially on dot products, so the 48 swaps give 24 reflections. The images of a cell under the boundary transformations and reflections that stay in the all-non-positive orthant of $S^6$ are its reduced presentations, the objects of the closure defined below.

### *2.3. Delone sorts and the root invariant*

Delone (published as Delaunay, 1933) classified lattices into 24 sorts by the pattern of zeros and equalities among the reduced conorms. Each sort lies in one Bravais type, and some Bravais types comprise several sorts (Delone *et al.*, 1975; Burzlaff & Zimmermann, 1985, 2006). We use the symbols C1 to A3 and the patterns of Andrews & Bernstein (2023) and SELLA (Andrews *et al.*, 2023), read in the slot order of Section 2.1; the correspondence with their slot order is given in the supporting information. The sorts are the Selling counterpart of the 44 Niggli lattice characters (Niggli, 1928; de Wolff & Gruber, 1991). They split a Bravais type by the shape of the reduced cell, not by its symmetry. A zero conorm restricts the metric without adding a symmetry: one right angle between two superbase vectors gives no twofold, so the triclinic sorts A2 and A3, with one and two zeros, are as triclinic as A1.

All obtuse superbases of a lattice carry the same six conorm values as a multiset, since a flip at a zero conorm only exchanges two slots. Which slots hold which value is not invariant: in the monoclinic example the flip at $ab$ moves $p_{ac}$ into the slot $da$. Kurlin (2022) normalises the square roots of the conorms, by a rule that depends on the Voronoi type, to the root invariant, a complete isometry invariant of the lattice. The invariant identifies a lattice; the closure records how its reduced cells are related.

### *2.4. Terms*

Table 2 collects the terms used in the rest of the paper, with the section in which each is defined. Strains are given in per cent throughout; $\epsilon = 7.4\%$ is 0.074 in the formulas.

Table 2. *Terms and symbols.*

| term | meaning | defined |
|---|---|---|
| superbase, conorm | four lattice vectors summing to zero; the six negated pair products $p_{ij}$ | 2.1 |
| obtuse, reduced cell | all six conorms $\geq 0$; an obtuse superbase, in some labelling | 2.1 |
| zero pattern, face | the slots whose conorms vanish; the conorm vectors $p \geq 0$ with those zeros | 2.1 |
| frame | the basis in which a metric tensor or superbase is written | 2.1 |
| flip, swap | the operators of Table 1; the 48 relabellings of a superbase | 2.2 |
| sort | one of Delone's 24 classes of lattices by zeros and equalities among the conorms | 2.3 |
| object, arrow | a labelled obtuse superbase (its metric tensor $\mathbf{G}$); a unimodular $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}'$ | 3.1 |
| visible symmetry, shape | a symmetry that relabels a reduced cell; the superbases whose conorms agree up to a swap | 3.1 |
| $\mathcal{U}$ | the 6096 integer matrices that are arrows of some closure | 3.3 |
| $e$ | the empty flip word, the reduced cell itself | 3.2 |
| strain | $\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F$, equation (2) | 4.1 |
| band $\epsilon$ | the strain within which a superbase or an operator is accepted | 4.2 |
| depth, price | strain from the measured metric to the nearest one that makes a superbase obtuse; strain by which an operator fails to carry one metric onto another | 4.2 |
| widened closure $\mathcal{S}_\epsilon$, $H_\epsilon$ | the superbases of depth $\leq \epsilon$; the self-maps of a cell of price $\leq \epsilon$ | 4.2 |
| parity class, $\mu_c$ | a class of $\Lambda/2\Lambda$ (lattice vectors modulo twice the lattice); the squared length of the shortest vector of class $c$ | 4.2, App. A |
| layer $\mathcal{N}_k$ | $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$ | 4.3 |
| adjacency radius $\delta_k$ | the strain at which a superbase outside $\mathcal{N}_k$ first becomes obtuse; computed up to a cap of 25% | 4.3 |
| difference-wall member | a member of the widened set reachable from the reduced cell only through a non-member, across the wall between two small conorms | 4.3 |
| sort spectrum | the 24 strain distances of a cell to the sorts | 4.4 |
| indexing class | a class of the Laue group (point group for anomalous data) among the candidates | 4.5 |
| price spectrum | the prices of the self-maps of a cell | 4.5 |

## 3. The Selling closure

### *3.1. Definition*

A lattice has no single preferred reduced cell. It has a finite set of obtuse superbases, and two kinds of integer matrix relate them: a symmetry maps a setting to itself and leaves its metric tensor unchanged, and a reindexing maps one setting to another, rewriting the metric tensor in the new basis without changing the lattice. A group holds only the first kind. A *groupoid* holds both: it is a collection of objects with

invertible arrows between them, in which two arrows can be composed only when the first ends where the second starts (Brandt, 1927; Higgins, 1971).

The group $GL_3(\mathbb{Z})$ acts on the right on positive definite symmetric matrices by $\mathbf{G} \cdot \mathbf{P} = \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$. Its action groupoid has an arrow $(\mathbf{G}, \mathbf{P})$ from $\mathbf{G}$ to $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$ for every $\mathbf{G}$ and every unimodular $\mathbf{P}$, and $(\mathbf{G}, \mathbf{P})$ followed by $(\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}, \mathbf{Q})$ is $(\mathbf{G}, \mathbf{P}\mathbf{Q})$. A *labelled* obtuse superbase is an obtuse superbase together with a choice of which three of its vectors, in which order and with which common sign, are called $a, b, c$; each obtuse superbase has 48 labellings, joined by swaps. The *Selling closure* $\mathcal{C}(\mathbf{G})$ of a lattice is the full subgroupoid of that action groupoid on the metric tensors of the lattice in its labelled obtuse bases: its objects are those finitely many tensors, and its arrows are all arrows of the action groupoid between them. The point group acts freely on labelled superbases, since a symmetry that fixes a labelled basis is the identity, and two labellings give the same metric tensor exactly when a symmetry carries one to the other; a lattice with $n$ obtuse superbases therefore has $48n/|\text{point group}|$ objects. In counting, the rest of the paper speaks of the $n$ unlabelled *superbases* of the closure. Three facts organise the closure.

1. *The arrows from an object to itself are the point group.* $\{\mathbf{P} : \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}\}$ is the lattice point group written in that basis (Himes & Mighell, 1987).

2. *The arrows between two objects are one coset of the point group.* If $\mathbf{P}_0$ is one arrow from $\mathbf{G}$ to $\mathbf{G}'$, every arrow between them is $g\mathbf{P}_0$ with $g$ a symmetry of $\mathbf{G}$, and equally $\mathbf{P}_0 g'$ with $g'$ a symmetry of $\mathbf{G}'$ (Flack, 1987).

3. *The closure is finite, and its arrows are known in advance.* A lattice has $n = 1$, 2, 3, 6 or 16 obtuse superbases according to the zero pattern (Kurlin, 2022; Lemma A1 gives the counts independently). Every arrow is a word of flips at zero conorms followed by a swap (Lemma A3), and every arrow belongs to one

fixed set $\mathcal{U}$ of integer matrices (Section 3.3).

The closure is connected, and a connected groupoid is equivalent, as a category, to the group of arrows at any one of its objects; facts 1 and 2 hold for any connected groupoid. What is particular to the closure is how the point group acts on the $n$ unlabelled superbases, on which $-\mathbf{I}$ acts trivially. The stabiliser of a superbase $F_0$ is the set of symmetries that merely relabel it, its *visible* symmetries. Its orbit is the set of superbases of the same *shape*, those whose conorm vectors agree with $F_0$'s up to a swap: a symmetry preserves $\mathbf{G}$ and hence the conorms, and conversely the relabelling that matches two equal conorm vectors is an arrow $\mathbf{P}$ with $\mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P} = \mathbf{G}$, a symmetry. By the orbit–stabiliser theorem the order of the point group is the number of visible symmetries times the number of superbases of $F_0$'s shape, and the number of shapes is the number of orbits. Every symmetry that is not visible carries the reduced cell to a different reduced cell, and no symmetry relates cells of different shape. A reduced cell here is always an obtuse superbase; a Niggli cell of type II (no acute angle) is one, and a Niggli cell of type I (three acute angles) is not.

### *3.2. One monoclinic cell*

Take $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with conorms

$$p_{da} = 639.5, \quad p_{db} = 1600.0, \quad p_{dc} = 2239.5, \quad p_{ab} = 0, \quad p_{ac} = 260.5, \quad p_{bc} = 0 \text{ Å}^2.$$

The two zeros are the right angles $\alpha$ and $\gamma$ at the twofold axis $b$, present in every primitive monoclinic lattice. The closure has six superbases (Table 3). Each has two zero conorms, each flip at a zero leads to a neighbour, and the six form a ring $M_0$–$M_1$–$M_5$–$M_4$–$M_3$–$M_2$–$M_0$ (Fig. 1). They fall into three shapes. $\{M_0, M_4\}$ is 30, 40, 50 Å, $\beta = 100°$; $\{M_1, M_3\}$ is 30, 40, 53.7 Å, $\beta = 113.4°$, the cell on $a$ and $a + c$; $\{M_2, M_5\}$ is

the third cell choice, on $c$ and $a + c$. These are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16).

Table 3. *The Selling closure of the monoclinic cell* $a = 30$, $b = 40$, $c = 50$ *Å*, $\beta = 100°$*: six obtuse superbases, each up to the order and common sign of its vectors, the flips at zero conorms that connect them, and where the four symmetries of the point group* $2/m$ *carry* $M_0$.

| object | four vectors | flips lead to | symmetries carry $M_0$ here |
|---|---|---|---|
| $M_0$ | $-a-b-c,\ a,\ b,\ c$ | $M_1$, $M_2$ | 1, $\bar{1}$ (swaps of $M_0$) |
| $M_1$ | $-b-c,\ -a,\ b,\ a+c$ | $M_0$, $M_5$ | |
| $M_2$ | $-a-c,\ a+b,\ -b,\ c$ | $M_0$, $M_3$ | |
| $M_3$ | $a+c,\ b-c,\ -b,\ -a$ | $M_2$, $M_4$ | |
| $M_4$ | $a-b+c,\ -c,\ b,\ -a$ | $M_3$, $M_5$ | 2 along $b$, $m$ normal to $b$ |
| $M_5$ | $a-b,\ c,\ b,\ -a-c$ | $M_1$, $M_4$ | |

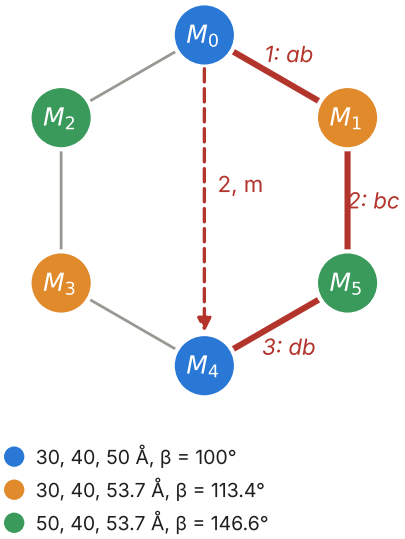



Fig. 1. The closure of the 100° cell: six superbases on a ring, joined by flips at the two zero conorms and coloured by cell shape. The red path is the flip word *ab.bc.db* from $M_0$ to $M_4$, each flip read in the labelling current at that step. The twofold and the mirror carry the vectors of $M_0$ to those of $M_4$ along this path, followed by a swap; they relabel no superbase on the way, so they are invisible in the reduced cell.

The point group is $2/m$, and the last column of Table 3 shows how its four operations act on the six superbases. The identity and the inversion are swaps of $M_0$: they relabel it. The twofold sends $(a, b, c) \mapsto (-a, b, -c)$ and so carries the four vectors $(-a-b-c, a, b, c)$ to $(a-b+c, -a, b, -c)$, which are the vectors of $M_4$, three flips away. As an

arrow of the closure each symmetry is a loop at the metric tensor of $M_0$; as a word in flips and swaps the twofold is the flips $ab$, $bc$, $db$ — the path $M_0 \to M_1 \to M_5 \to M_4$ — followed by the swap that renames the vectors of $M_4$ so that its metric tensor is that of $M_0$, and the mirror is the same flips with that swap composed with the change of common sign. Half of the point group is therefore invisible in the reduced cell: the stabiliser of $M_0$ has two elements, and the orbit of $M_0$ is $\{M_0, M_4\}$, as the count $4 = 2 \times 2$ requires. "Visible" means only that an operation relabels the reduced superbase; a search over lattice directions, such as Le Page's, finds the twofold either way. Two measurements that reduce to $M_0$ and to $M_1$ report different cell parameters and are related by no symmetry, only by the flip at $ab$.

### *3.3. The universal operator set*

The arrows can be characterised with no lattice at all. An integer matrix $\mathbf{W}$ acts on the six conorms by a fixed integer $6 \times 6$ matrix, and $\mathbf{W}$ is an arrow of the closure of every lattice on a face exactly when that matrix maps the face into the cone of non-negative conorms, a sign condition on its entries (supporting information, Section S3). The union over the 38 faces is a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices, with entries in $\{-3, \ldots, 3\}$ and closed under inversion. By Lemma A1 it contains every arrow between two reduced cells of any lattice. The superbases $\mathcal{U}F_0$, 127 up to relabelling, are all the superbases that can be obtuse together with $F_0$ for some metric. This set carries the completeness statements of Section 4.

## 4. Measured cells: methods

A measured cell is never exactly on a zero conorm: a right angle is measured as 90.25°, and the conorm is a small number of either sign. The flips that joined the superbases of Table 3 now lead to superbases with a slightly negative conorm, Selling reduction

rejects them, and the exact closure of the measured metric has one superbase instead of six. The twofold and the mirror, which were arrows to $M_4$, now lead outside the set. The closure has to be widened.

### *4.1. The strain norm*

The measure used is the relative change of the metric tensor,

$$\|\Delta\mathbf{G}\|_{\mathbf{G}} = \|\mathbf{G}^{-1/2}\,\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F, \tag{2}$$

a strain. It is the norm of the affine-invariant Riemannian metric on positive definite matrices at $\mathbf{G}$ (Pennec *et al.*, 2006), and so the geodesic distance to first order in $\Delta\mathbf{G}$. It is unchanged by a change of basis, since $\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\mathbf{G}\mathbf{P}$ and $\Delta\mathbf{G} \mapsto \mathbf{P}^{\mathsf{T}}\Delta\mathbf{G}\mathbf{P}$ leave it as it is: it is the same in every frame, every arrow of the closure is an isometry for it, and it is unchanged by a common scale of the cell. A perturbation of norm $\epsilon$ changes the dot product of any two lattice vectors by at most $\epsilon|u||v|$ (Proposition A4), so a squared edge by at most a fraction $\epsilon$ and, to first order, the cosine of any angle by at most $\epsilon$, whatever the shape of the cell. An absolute tolerance on the conorms in Å$^2$, as is implicit in a tolerant reduction, behaves differently for a cell whose edges differ in length: for a needle the error on the conorm between the two long vectors is hundreds of times that between the short ones, and the frame in which a conorm is read changes its tolerance. That band is a different norm, an $L^\infty$ band on the conorms, comparable with the strain only for cells of similar edge lengths; its theory is in the supporting information (Section S4).

The norm is a local one: it is measured at a metric, and two metrics give the same value only to first order. Prices and distances in this paper are measured at the metric named, the statements that compare values at different metrics carry an explicit factor (Lemma A8, Proposition A10), and a value of 25% is a bound in this linearised norm.

*Tolerances in this norm.* Table 4 converts the classical tolerances. Le Page's search

(Le Page, 1982) accepts a twofold whose obliquity $\delta$, the angle between a lattice row and the nearest reciprocal row, is at most 3°. For one twofold the distance from the cell to the metrics that the twofold fixes is $\sqrt{2}\sin\delta$ to first order, 7.4% at 3°, and the price of the twofold is twice that, 14.8% (Proposition A9). A sort that needs several twofolds is further away than any one of them, because Le Page takes the largest obliquity and the sort distance combines them: a $30 \times 40 \times 50$ Å cell with $\alpha = \beta = \gamma = 92°$ has obliquity 2.88° on each axis, so a 3° search accepts oP and each twofold has a price of 14.3%, but its distance to oP is 8.85%. The conversion of 3° to a sort distance of 7.4% therefore holds only for sorts reached by one twofold; the conversion to a price of 14.8% holds for every twofold separately. The norm weighs all directions of the metric alike; Section 4.5 sketches the Mahalanobis form that replaces it when a covariance of the metric is known.

Table 4. *Classical tolerances as strains (equation 2). Each value is to first order; for an orthogonal pair the angle conversion is exact.*

| tolerance | strain | at 3° or 5% |
|---|---|---|
| angle $\theta$ between two vectors changed by $\delta$ | $\delta\sqrt{2(1+\cos^2\theta)}/\sin\theta$; $\sqrt{2}\sin\delta$ at 90° | 7.4% |
| edge $a$ changed by a fraction $f$ | $2f\,\lvert a\rvert^2\lvert a^*\rvert^2$; $2f$ for an axis normal to the other two | 10% |
| one twofold of obliquity $\delta$: distance to its fixed metrics | $\sqrt{2}\sin\delta$ | 7.4% |
| one twofold of obliquity $\delta$: its price a sort needing several twofolds | $2\sqrt{2}\sin\delta$ at least the largest of their distances | 14.8% |

### *4.2. The widened set*

The *widened closure* of a metric $\mathbf{G}$ at band $\epsilon$ is

$$\mathcal{S}_\epsilon(\mathbf{G}) = \big\{\, F \;:\; F \text{ is obtuse for some metric within } \epsilon \text{ of } \mathbf{G} \text{ in the norm (2)} \,\big\}. \qquad (3)$$

Its superbases are the reduced cells of every lattice consistent with the measurement at band $\epsilon$, and its arrows the unimodular matrices between their labellings. Each superbase has a *depth*, the strain from the measured metric to the nearest metric that

makes it obtuse, a non-negative least-squares problem in its six conorms; it belongs to $\mathcal{S}_\epsilon(\mathbf{G})$ when its depth is at most $\epsilon$. Each arrow $\mathbf{P}$ from $\mathbf{G}_1$ to $\mathbf{G}_2$ has a *price*, the strain $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}_1\mathbf{P} - \mathbf{G}_2\|_{\mathbf{G}_2}$ by which it fails to carry one metric onto the other, zero for every arrow of an exact closure. The self-maps of price at most $\epsilon$ of the measured cell, written $H_\epsilon$, take the place of the point group; whether they form a group is settled by Proposition A10. Equation (3) names no starting cell, order of operations or frame, so $\mathcal{S}_\epsilon(\mathbf{G})$ is the same whoever computes it. Two properties make it computable and useful.

*It is finite.* The lattice vectors modulo twice the lattice fall into eight *parity classes*; seven of them do not contain the zero vector, and the seven partial sums of an obtuse superbase, the sums of one or two of its vectors up to sign, are shortest vectors of those classes. Write $\mu_c$ for the squared length of the shortest vector of class $c$. Every vector of a superbase in $\mathcal{S}_\epsilon(\mathbf{G})$ lies in a ball of squared radius $\frac{1+\epsilon}{1-\epsilon}\mu_c$ about the origin (Lemma A6). The lattice vectors in these seven balls are finitely many, so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly, with no window of coordinates to assume.

*Widening one side suffices.* If a new cell, carried to a reference by an operator $\mathbf{P}$, differs from it by a price of at most $\epsilon$ measured at the new cell, then the new cell's reduced superbase, carried by $\mathbf{P}$, lies in the reference's widened closure at band $\epsilon/(1-\epsilon)$ (Lemma A8). So only the reference is widened, by that slightly larger band, and the new cell is reduced once.

### *4.3. Computing the widened set*

For a measured cell and a band $\epsilon < 1$:

1. Selling-reduce the measured cell to a superbase $F_0$.

2. Enumerate the lattice vectors in the seven balls of Lemma A6 and form the superbases they make.

3. Keep the superbases of depth at most $\epsilon$. Record each one's depth and the operator $\mathbf{W}$ from $F_0$ to it.

This is exact for every cell and every band below 1, and it is what the implementation does.

*Why enumerate.* Reduction with a widened boundary does not stop at a well-defined cell, and walking from the reduced cell, flipping wherever a conorm is within a band of zero, does not reach every member. A member can be reachable only through a non-member: with two small conorms at 1.2 and 0.5 times the band, the flip across the first leaves the band and only a second flip, across the other, returns into it. We call such a member a *difference-wall member*, since it lies across the wall where the two small conorms are equal. It carries no symmetry; it is an ordinary alternative reduced cell to which a second measurement can reduce, and a reference set built by walking never contains it. Section 5.2 and the supporting information (Table S3) report how often walks fail.

*Layers.* Call two superbases *adjacent* when some metric makes both obtuse. The superbases adjacent to $F_0$ are the first layer $\mathcal{N}_1 = \mathcal{U}F_0$, and $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$. The *adjacency radius* $\delta_k(\mathbf{G})$ is the strain from $\mathbf{G}$ to the nearest metric for which a superbase outside $\mathcal{N}_k$ is obtuse; the implementation computes it up to a cap of 25%. If $\epsilon < \delta_k$, the widened set lies in the $k$th layer and is the set of its members of depth at most $\epsilon$ (Remark A7). The *layer* of a cell at band $\epsilon$ is the least $k$ for which $\mathcal{S}_\epsilon(\mathbf{G}) \subseteq \mathcal{N}_k$. Computing $\delta_k$ costs the same ball enumeration as computing the set, so the layers do not give a cheaper certified answer; they describe the answer, and the first layer alone, 127 depths, is a cheap route that is exact whenever $\epsilon < \delta_1$. No lower bound on $\delta_1$ in terms of the cell's geometry is proved here.

*Projection.* The set can also be built from the lattices near the measurement. For each of the 127 frames of the first layer and each of the 38 faces, project the measured

metric onto the face (the nearest metric in the norm (2) with those conorms zero and the others non-negative); for every projection within $\epsilon$, enumerate the exact closure of the projected lattice; keep the members of depth at most $\epsilon$; take the union. Below $\delta_1$ this returns $\mathcal{S}_\epsilon(\mathbf{G})$ (Remark A7). A single projection, onto one face in one frame, is not enough (Section 5.2).

*Ties take no band.* Only zeros enter the construction. Every sort lies on one zero pattern, and every symmetry of every lattice on that pattern is one of its arrows (Lemma A3), so a near-equality — the $a \approx b$ of a pseudo-tetragonal cell — never adds a superbase: the operator that would be its symmetry is already enumerated, and whether it is accepted is decided by its price.

### *4.4. Distance to each sort*

The same norm measures how far a measured cell is from each of the 24 sorts. In each of the 6096 frames of $\mathcal{U}$, the cell's conorms are projected onto the sort's subspace (its zeros and equalities, Proposition A5, the other conorms non-negative), and the smallest distance is taken. This is the computation of SELLA (Andrews *et al.*, 2023), which projects the boundary images of a cell in $S^6$ onto the sorts' subspaces, and of Oishi-Tomiyasu (2012) in Selling coordinates; projection onto Bravais-lattice subspaces goes back to Andrews & Bernstein (1988) in $G^6$. What changes here is the norm, which is the same in every frame, and the set of frames, which is $\mathcal{U}$. We call the 24 distances the *sort spectrum* of the cell. A distance below the adjacency radius $\delta_1$ is exact, since the frames of $\mathcal{U}$ contain the frame of the nearest lattice of the sort (Remark A7). A minimum above $\delta_1$ is only an upper bound over the frames used, and the true distance is at least $\delta_1$; the implementation flags such entries. The *step* between two sorts joined by a specialisation (supporting information, Table S2) is the distance from the cell idealised onto the upper sort to the lower one.

*4.5. Reindexing*

A reference $R$ is stored as its widened closure, the operators $\mathbf{W}_k$ from its reduced cell to the members. A new cell $N$ is reduced once, to a basis $B_N$ in its own coordinates; $B_R$ is the reduction matrix of the reference. For each stored $\mathbf{W}_k$ and each of the 48 swaps $\mathbf{Q}$ the candidate operator from the reference's input basis to the new cell's is

$$\mathbf{P} = B_R \mathbf{W}_k \mathbf{Q}\, B_N^{-1}, \tag{4}$$

so that $(a_N\, b_N\, c_N) = (a_R\, b_R\, c_R)\mathbf{P}$, $\mathbf{G}_N \approx \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$, and Miller indices transform as $h_N = \mathbf{P}^{\mathsf{T}} h_R$. Only proper operators, of determinant $+1$, are kept: a left-handed basis is never a valid output, whatever the space group. Each candidate is priced by $\|s\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P} - \mathbf{G}_N\|_{\mathbf{G}_N}$, with $s$ the common scale that minimises that norm, since a detector-distance or wavelength error scales every conorm alike; the fitted $s$ is reported with each candidate, and a value far from 1 signals such an error rather than passing silently. Candidates of price at most $\epsilon$ are returned. For the true operator to be among them whenever its price is at most $\epsilon$, the reference is widened at $\epsilon/(1-\epsilon)$ (Lemma A8), 8.0% for $\epsilon = 7.4\%$. A centred reference is first taken to a primitive cell, and the centring matrix is composed with the operator found, which is then rational in the conventional basis; carrying the result to the conventional setting of the target space group is done outside the closure.

*Cosets and classes.* Let $H_\epsilon$ be the unimodular self-maps of the reference of price at most $\epsilon$, a finite set, and $p_0$ the price of the true operator $\mathbf{P}_0$. Every candidate is $h\mathbf{P}_0$ with $h$ a self-map of price at most $(\epsilon + p_0)/(1 - p_0)$, and every $h$ of price at most $(\epsilon - p_0)/(1 + p_0)$ gives a candidate (Proposition A10, exact for prices without the scale fit). If no self-map has a price in $(\epsilon, 2\epsilon + \epsilon^2]$, then $H_\epsilon$ is a group; if in addition none has a price between the two bounds just given, the candidates are exactly the coset $H_\epsilon\mathbf{P}_0$. The implementation checks closure of $H_\epsilon$ directly. Two candidates that differ

on the left by an element of the reference's Laue group $\mathcal{L}$ index the crystal the same way, so the distinct indexings are the classes of $\mathcal{L}$ among the candidates, $[H_\epsilon : \mathcal{L}]$ of them when the coset statement holds. With proper operators only, the Laue group enters through its proper rotations. For anomalous data from an acentric crystal the point group replaces the Laue group, since the inversion then changes the indexing. The guarantee is that the true operator is among the candidates; the price orders the classes, but it does not single out the true member of a class, which is a matter for the intensities, as in a twinning test.

*Which pseudo-symmetries are offered.* A symmetry $g$ of a nearby lattice enters $H_\epsilon$ when its price is at most $\epsilon$. For a twofold the price is exactly twice the distance from the measured metric to the metrics $V_g$ that $g$ leaves invariant, in the geodesic distance, and in the norm (2) at a metric of $V_g$; for a rotation of order $n$ the ratio lies between 1 and 2 ($\sqrt{3}$ for a threefold, between $\sqrt{2}$ and 2 for a fourfold, between 1 and $\sqrt{3}$ for a sixfold; Proposition A9). The subspace of a sort that $g$ belongs to lies inside $V_g$, so the price is at most twice the distance to that sort, with equality when $V_g$ is the sort's subspace, as for the twofold of a primitive monoclinic lattice. A twofold whose fixed metrics are within $\epsilon/2$ of the measurement is therefore offered: at $\epsilon = 7.4\%$, a pseudo-twofold of obliquity up to about 1.5°; at 14.8%, every twofold that Le Page's 3° accepts.

*Choosing the band.* The band is a tolerance on how much two measurements of one crystal may differ. Two sources set it. The first is measurement error. For two independent measurements of one lattice, each with isotropic strain noise of root-mean-square $\sigma$, the price of the true operator has median $1.2\sigma$ and 99th percentile $2.2\sigma$, and the price of an exact twofold in one measurement has median $1.0\sigma$ and 99th percentile $2.5\sigma$; a band of $2.5\sigma$ therefore keeps the true operator and the exact symmetries. The $\sigma$ of a cell follows from its standard uncertainties by first-order propagation through

$\mathbf{G}(a, b, c, \alpha, \beta, \gamma)$ (`strain_from_esds` in the implementation). The second source is non-isomorphism: cells of one macromolecular crystal form commonly differ by 0.5 to 2% in their edges, a strain of 1 to 4%, which is added to the band. Le Page's angular tolerance is a different kind of number, a tolerance on how far a lattice may be from a symmetry it is credited with, and enters through Proposition A9. When a covariance $\Sigma$ of the six components of $\Delta\mathbf{G}$ is known, the Mahalanobis norm $\Delta\mathbf{G}^{\mathsf{T}}\Sigma^{-1}\Delta\mathbf{G}$ replaces equation (2). It stays basis-invariant when $\Sigma$ transforms with the metric, and every metric within a Mahalanobis distance $\epsilon_\Sigma$ is within a strain $s_{\max}\epsilon_\Sigma$, with $s^2_{\max}$ the largest eigenvalue of $\Sigma$ written in strain coordinates, so the strain enumeration at that band followed by a Mahalanobis filter is exact. That extension is not implemented here.

### *4.6. Verification, and what is proved*

Exact closures and the closures of the 24 sorts were checked against brute force over every basis with entries in $[-2, 2]$ relative to $F_0$. That window suffices: every member of every exact closure, found over all 38 faces in the window of Lemma A1 with no bound on the entries, has a labelling with entries in $[-2, 2]$. Widened sets were checked against the ball enumeration of Lemma A6, which needs no window, and against brute force over $[-3, 3]$ for needles and plates. The measured worked cell is 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80°: the 100° cell of Section 3.2 with $\beta$ moved to 92°, so that it sits near two zero conorms and not far from a third, perturbed by a metric error of strain 0.82%. Table 5 separates what is proved from what is computed. The proofs rest on Selling's identity, the characterisation of obtuse superbases by Voronoi vectors (Conway & Sloane, 1992; at a zero conorm a partial sum is a Voronoi vector that is not strict) and Kurlin's enumeration, whose counts are reproduced here independently.

Table 5. *What is proved, what rests on an exhaustive computation, and what is checked or refuted.*

| statement | status |
|---|---|
| every arrow is flips at zero conorms and a swap; the members of each closure lie in a finite window | proved (Lemmas A1, A3) |
| $\mathcal{U}$ contains every arrow between reduced cells of any lattice | computer-assisted: the window of Lemma A1 enumerated over all 38 faces |
| the 24 sorts | a definition (Proposition A5) and an enumeration, matched to Delone's patterns by computation |
| the strain norm bounds every dot product | proved (Proposition A4) |
| the widened set is finite and enumerated exactly | proved (Lemma A6) |
| the filtered layer $\mathcal{N}_k$ is the widened set below $\delta_k$; projection is exact below $\delta_1$ | follows from the definition of $\delta_k$ and from $\mathcal{U}$ (Remark A7); the radii are computed per cell |
| widening the reference alone suffices, at $\epsilon/(1-\epsilon)$ | proved (Lemma A8) |
| the price of a twofold is twice its distance to the metrics it fixes; ratios for order $n$ | proved: exact in the geodesic distance and in the norm at a fixed metric (Proposition A9) |
| the candidates lie between two cosets of self-maps; $H_\epsilon$ is a group in a gap of the price spectrum | proved, for prices without the scale fit (Proposition A10) |
| the layers of 127 and 2755 superbases; the counts of Lemma A1 and Proposition A5; the adjacency radii | computer-assisted enumeration |
| projection is exact between $\delta_1$ and $\delta_2$ | checked on every case tested |
| $\delta_1$ between 1 and $\sqrt{2}$ times the ratio of shortest to longest edge | observed on four cells; a conjecture |
| bands per slot applied in the frame of the reduced cell | refuted (the needle of Section 5.2) |

## 5. Results

### *5.1. The closures of the 24 sorts*

For each face the arrows are the operators of $\mathcal{U}$ that map the face into the cone; on the face every arrow permutes the six conorm slots, so each fixes a subspace of conorm vectors, and the sorts are the closed sets of equalities obtained by intersecting those subspaces (Proposition A5). For a sort, the point group is the set of arrows that fix the sort's subspace pointwise, the visible symmetries are those that are swaps of the reduced cell, and two superbases have the same shape when their conorms agree up to a slot permutation. Every entry of Table 6 is the result of this linear algebra on integer matrices. In every sort the number of symmetries out of each labelled superbase is the order of the holohedry: 48, 24, 16, 12, 8, 4 or 2. The twofolds, with the inversion,

generate the group in every sort, so finding the twofolds suffices.

Table 6. *The Selling closure of each Delone sort, computed from the operators alone. Patterns are conorms in the slot order $da, db, dc, ab, ac, bc$; equal letters denote equal values and* 0 *a zero; they are the patterns of Andrews & Bernstein (2023) read in this order. Superbases are counted up to swaps. "Visible" is the number of symmetries that relabel the reduced superbase written in the pattern shown; another superbase of the same lattice can show a different number. "Shapes" is the number of distinct cell shapes among the superbases, equal in every case to the number of orbits of superbases under the symmetries.*

| sort | Bravais | pattern | zeros | superbases | \|group\| | visible | shapes |
|---|---|---|---|---|---|---|---|
| C1 | cI | rrrrrr | 0 | 1 | 48 | 48 | 1 |
| C3 | cF | rr00rr | 2 | 3 | 48 | 16 | 1 |
| C5 | cP | rrr000 | 3 | 16 | 48 | 12 | 2 |
| H4 | hP | rrsr00 | 2 | 6 | 24 | 4 | 1 |
| T1 | tI | rrssrr | 0 | 1 | 16 | 16 | 1 |
| T2 | tI | rrs0rr | 1 | 2 | 16 | 8 | 1 |
| T5 | tP | rrs000 | 3 | 16 | 16 | 4 | 3 |
| R1 | hR | sssrrr | 0 | 1 | 12 | 12 | 1 |
| R3 | hR | sr00rr | 2 | 3 | 12 | 4 | 1 |
| O1A | oF | rrtsrr | 0 | 1 | 8 | 8 | 1 |
| O1B | oI | rsttsr | 0 | 1 | 8 | 8 | 1 |
| O2 | oI | srt0sr | 1 | 2 | 8 | 4 | 1 |
| O3 | oI | rs00sr | 2 | 3 | 8 | 8 | 2 |
| O4 | oS | sstr00 | 2 | 6 | 8 | 4 | 2 |
| O5 | oP | rst000 | 3 | 16 | 8 | 2 | 4 |
| M1A | mC | ttusrr | 0 | 1 | 4 | 4 | 1 |
| M1B | mC | rsutsr | 0 | 1 | 4 | 4 | 1 |
| M2A | mC | stu0sr | 1 | 2 | 4 | 2 | 1 |
| M2B | mC | rst0sr | 1 | 2 | 4 | 4 | 2 |
| M3 | mC | ts00sr | 2 | 3 | 4 | 4 | 2 |
| M4 | mP | stur00 | 2 | 6 | 4 | 2 | 3 |
| A1 | aP | uvwtsr | 0 | 1 | 2 | 2 | 1 |
| A2 | aP | tuv0sr | 1 | 2 | 2 | 2 | 2 |
| A3 | aP | tu00sr | 2 | 3 | 2 | 2 | 3 |

*Zero pattern and sort fix different parts.* Which superbases exist and which arrows join them is fixed by the zero pattern; which arrows are symmetries is fixed by the sort, and the abstract group by the Bravais type. Primitive monoclinic, C-centred orthorhombic and primitive hexagonal lattices all have two zeros sharing a vector and six superbases, and differ only in how many arrows are symmetries (4, 8, 24). Within one sort the shape of the cell plays no part: the superbases, arrows and symmetries are the same, and the lattice supplies only the numbers attached to them.

*Symmetry visible in one cell.* In the sorts with zero conorms the symmetries that relabel the reduced cell are a minority: 2 of 8 for primitive orthorhombic, 4 of 16

for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal, 4 of 12 for rhombohedral R3. None of the three twofolds of a primitive orthorhombic lattice relabels its reduced cell, nor the one twofold of primitive monoclinic; of the five, nine and seven twofolds of primitive tetragonal, cubic and hexagonal lattices, one, three and one do.

*Cell shapes.* The orbit of the reduced cell under the point group has |group|/visible superbases, all of its shape. For cP the orbit of the orthogonal cell has $48/12 = 4$ superbases, the four orthogonal superbases, and the twelve others form the second shape; for tP it has $16/4 = 4$, for oP $8/2 = 4$, and for hP all $24/4 = 6$ superbases are one shape. Ten of the 24 sorts have more than one shape. Primitive orthorhombic has four: the orthogonal cell and three monoclinic-looking cells with two right angles, one for each axis traded for a face diagonal. Reduction of a measured orthorhombic cell stops at whichever of the sixteen superbases its errors favour, and the orthogonal cell is up to three flips away, reached by no twofold of the lattice. The catalogue of shapes is Table S1 of the supporting information.

*Specialisations.* The same algebra gives the conditions under which a lattice of one sort becomes a lattice of another (supporting information, Table S2). Every condition of codimension one is a conorm vanishing or two conorms becoming equal; they join 36 pairs of sorts. The classical values $c/a = \sqrt{2}$ (tI to cF), $b = \sqrt{3}\,a$ (oS to hP) and $a = c$ in a monoclinic cell are among them, and the monoclinic conditions come in threes, one per cell choice. The two kinds behave differently under error. A near-equality, such as $c/a \approx \sqrt{2}$, changes no zero pattern: the set of superbases is already complete, and the pseudo-symmetry appears as a priced arrow. A near-zero, such as a C-centred monoclinic cell one zero from primitive tetragonal, adds superbases, and those are what the widened closure recovers.

*5.2. Completeness on measured cells*

*The worked cell.* The widened closure of the worked cell grows with the band: one member at $\epsilon = 0.1\%$, two at 0.3%, three at 0.6%, the six of the ring from 1%, and sixteen at 7.4% (Fig. 2; the members, as flip words with their conorms and depths, are listed in Table S6 of the supporting information). The six members of the ring enter below 0.8%; the other ten members of the closure of the nearby orthorhombic lattice enter between 3.8 and 5.1%. Depths are not ordered as the most negative conorms are, because the strain weighs each conorm by the lengths of the vectors it joins.

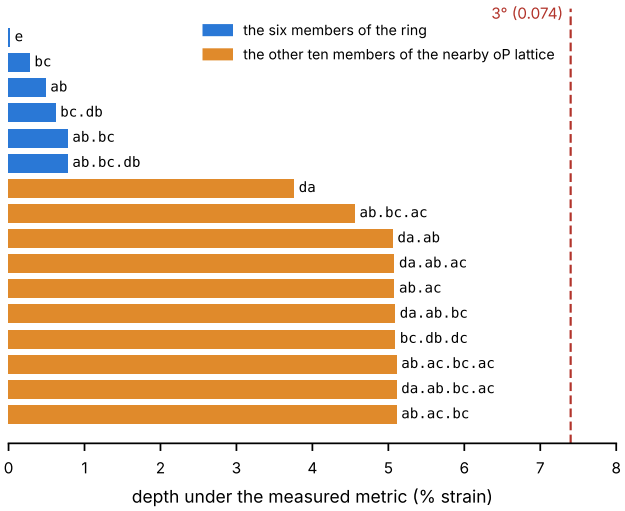



Fig. 2. The sixteen members of the widened closure of the measured worked cell by depth: the six of the ring enter below 0.8% strain, the other ten members of the nearby orthorhombic lattice's closure between 3.8 and 5.1%; the band selects among them.

*Walks miss members.* With a band on the conorms, reduction of the worked cell and of a noisy orthorhombic cell from random starting bases stopped at several different members of the widened set, and on a noisy cubic cell the restarts did not find all of them. On 80 cells with zero conorms at awkward positions, four versions of the walk of Section 4.3 returned a wrong set on 2 to 13 cells (Table S3). The cause in each case is a difference-wall member.

*Needles and plates.* Bands per slot, $\sigma|u_i||u_j|$, applied in the frame of the reduced cell, are not exact: the needle $5.58 \times 5.21 \times 131.97\,\text{Å}$, $\gamma = 91.12°$, at $\sigma = \sin 0.064°$ gives 6 superbases where the per-slot widened set has 12, because on the slot $(a+c, b)$ of the missing members the same dot product carries a much larger band. The strain norm has no frame-dependent band. The projection construction returned the exact set for this needle at bands of 0.05, 0.2, 0.5, 1, 1.7 and 5.2%; for a $100 \times 100.1 \times 5\,\text{Å}$ plate and a $150 \times 120 \times 4.8\,\text{Å}$ plate, $\beta = 90.4°$, at 5.2%, where their widened sets have 20 and 28 members, more than any single closure; and for 60 random needles, plates, isotropic and monoclinic cells at four bands each from 0.17 to 5.2%, in all 240 cases, against brute force. A single projection onto the most special face is not enough: on the needle at 0.2 and 0.5% it returns 6 of the 8 members, missing those built on $a+c$, which belong to a lattice in which $(a+c)\cdot b$, not $a \cdot b$, is zero.

*Adjacency radii.* Table 7 gives the adjacency radii. For the worked cell and the $10 \times 10 \times 30\,\text{Å}$ needle no superbase outside the first layer comes within the 25% cap, so filtering the first layer is exact at every band in use. For needles and plates $\delta_1$ falls with the ratio of the shortest to the longest edge, between one and $\sqrt{2}$ times that ratio in the four cells of the table with $\delta_1$ below the cap, and the second layer roughly doubles it. The two plates at 5.2% lie between $\delta_1$ and $\delta_2$: the first layer misses members there, and the second contains them all. Their $\delta_2$ also exceeds 8.0%, the band at which a reference is widened for $\epsilon = 7.4\%$, so at that band the filtered second layer is their widened set; at $\epsilon = 14.8\%$ the reference band of 17.4% exceeds $\delta_2$ for three of the four, and only the ball enumeration is certified. For the cells of the BLDConograph samples, the lysozyme cell of Section 5.4 and the worked cell, $\delta_1$ reaches the cap. In the benchmark of Section 5.3 the widened set of every reference of ordinary shape lay in the first layer, and of the 16 needles and plates 12 lay in the second and 4 in the first. On one core of a shared machine, widening a reference at 8.0% took 0.1 to 2 s

for cells of ordinary shape and 4 to 6 s for needles and plates, and reindexing one new cell 0.4 to 2 s.

Table 7. *Adjacency radii in the strain norm (equation 2), in per cent: $\delta_1$, the smallest strain at which a superbase outside the first layer becomes obtuse, and $\delta_2$ for the second layer. The search is capped at* 25%.

| cell | $\delta_1$ | $\delta_2$ |
|---|---|---|
| $30 \times 40 \times 50$ Å, $\beta = 92°$ (the worked cell) | $> 25$ | $> 25$ |
| needle $10 \times 10 \times 30$ Å | $> 25$ | $> 25$ |
| needle $10 \times 10 \times 100$ Å | 13.9 | $> 25$ |
| needle $5.58 \times 5.21 \times 132$ Å, $\gamma = 91.12°$ | 5.6 | 11.1 |
| plate $100 \times 100.1 \times 5$ Å | 5.0 | 14.1 |
| plate $150 \times 120 \times 4.8$ Å, $\beta = 90.4°$ | 4.3 | 10.0 |

### *5.3. Reindexing*

*Indexing classes.* Table 8 and Fig. 3 give the classes for six references, each against the same crystal measured again with a 0.1% error in the length scale and a relabelling of its axes, at $\epsilon = 7.4\%$. The scale fit absorbs the 0.1% error, so the exact symmetries cost nothing. For the exactly symmetric references the counts are the index of the Laue group in the lattice holohedry, the classical table of indexing ambiguities for merohedral crystals (Yeates, 1997; Dauter, 2003; Evans, 2006, 2011): tetragonal lysozyme in $P4_32_12$ has one class, $P3$ has four, joined by the twofolds along [001], [110] and $[1\bar{1}0]$, and $P3_121$ and $P2_13$ have two, joined by one merohedral twofold. This is a restatement in the strain norm, not a new result. What the price adds is the pseudo-merohedral classes: a pseudo-tetragonal $P2_12_12_1$ cell, $40.0 \times 40.3 \times 80$ Å, and the worked cell in $P1$ each have a second class, at a price of 2.11% and 1.55%, from the twofold that would make the cell tetragonal or monoclinic. That price is twice the cell's distance to the sort the twofold fixes, 1.06% and 0.78%, as Proposition A9 gives when the twofold's fixed metrics are the sort's subspace. The band decides whether such a class is offered: along a scan of $\beta$ for a monoclinic $30 \times 40 \times 50$ Å cell the pseudo-orthorhombic class is offered at $\epsilon = 7.4\%$ for $\beta = 91.4°$ and not for 91.6°, and at 14.8% for 92.5° and not for 93.5°, as the rule predicts (limits 91.5° and 93°). With

proper operators only, anomalous data change nothing for a chiral crystal: lysozyme keeps one class and $P3$ four. For an acentric crystal that is not chiral the inversion separates indexings when anomalous differences are used: $P4mm$ goes from one class to two and $P31m$ from two to four.

Table 8. *Distinct indexings of six references at* $\epsilon = 7.4\%$ *(the reference widened at* $8.0\%$*): the order of the band's group of proper self-maps* $H_\epsilon$ *and of the proper rotations of the Laue group* $\mathcal{L}$*, the number of classes of* $\mathcal{L}$ *among the candidates, and the price of each class. The new cell is the reference with a* $0.1\%$ *scale error, which the scale fit absorbs, relabelled.*

| reference | space group | $\lvert H_\epsilon \rvert$ | $\lvert \mathcal{L} \rvert$ | classes | prices (%) |
|---|---|---|---|---|---|
| $79.1 \times 79.1 \times 37.9$ Å | $P4_32_12$ | 8 | 8 | 1 | 0 |
| $a = b = 50$, $c = 80$ Å | $P3$ | 12 | 3 | 4 | 0, 0, 0, 0 |
| $a = b = 50$, $c = 80$ Å | $P3_121$ | 12 | 6 | 2 | 0, 0 |
| $a = 60$ Å | $P2_13$ | 24 | 12 | 2 | 0, 0 |
| $40.0 \times 40.3 \times 80$ Å | $P2_12_12_1$ | 8 | 4 | 2 | 0, 2.11 |
| worked cell | $P1$ | 2 | 1 | 2 | 0, 1.55 |

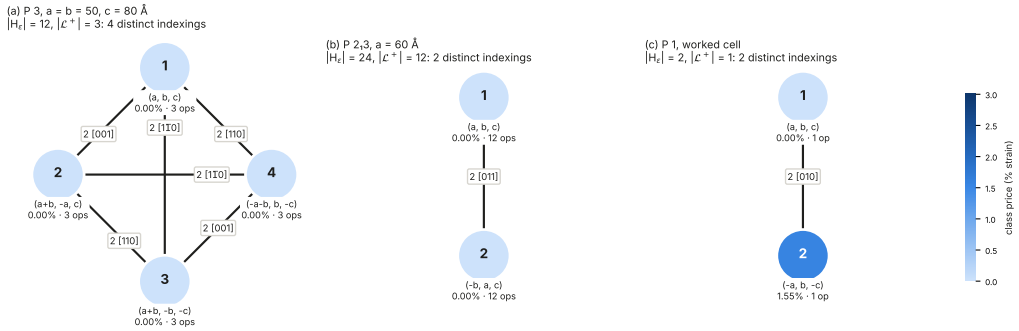



Fig. 3. Three of the cases of Table 8 as graphs: the classes of the Laue group among the candidates for (*a*) $P3$, (*b*) $P2_13$ and (*c*) the worked cell in $P1$. Each node is one class; it shows the class's operator nearest the identity (new basis in terms of the reference's $a$, $b$, $c$), its strain price and the number of operators in it. Each edge is the lattice symmetry or pseudo-symmetry carrying one class to the other, as rotation order and axis $[uvw]$ in the reference basis. Node colour is the price (colour bar).

*The benchmark.* The benchmark (supporting information, Section S2) has 780 constructed cases, primitive cells with a random metric error and a random proper reindexing, built where the closure predicts trouble: 75 cells measured in another shape of the same lattice, 660 lattices a small distance from a specialisation, 20 whose only reduced superbase is a difference-wall member, 16 needles and plates, and 9 cells of

the boundary sorts. The errors are small, a median strain of 0.1% against the band of 7.4%. A second family, the stress cases, puts independent errors on both cells, scaled so that the true operator's price is 0.5 and 0.9 times the band, on 60 references drawn from the benchmark; at 1.1 and 1.5 times the band, where no method should accept the true operator, none did. The third family is real: 210 cells from the Crystallography Open Database flagged by a census as lying near zero conorms or other places where reduction is ambiguous, each with errors drawn from its own deposited standard uncertainties, relabelled at random, at a band of 2.5 times its own error strain (median 0.88%).

*Outcome.* Table 9 scores six methods on the three families: the closure; a search over every unimodular matrix with entries in $\{-1, 0, 1\}$ (6960 matrices) or $\{-2, \ldots, 2\}$ between the two Selling-reduced cells, priced in the same norm; cctbx `similarity_transformations` between the two Niggli cells (Grosse-Kunstleve *et al.*, 2002) with generator ranges 1 and 2; and a pipeline that runs the cctbx lattice-symmetry analysis (Le Page's 3°; Sauter *et al.*, 2004) on both cells, takes the reference's highest lattice group and the same group for the new cell, and returns every operation of that group between the two standard settings. Each method's candidates are kept when their price is at most the band. The closure found the true operator in every case, as Lemma A8 guarantees. The small-matrix searches and the similarity transformations found it nearly as often; the pipeline missed cases in which the two cells' highest lattice groups differ at the tolerance. On the real cells every method found it. These numbers confirm the guarantee; they do not show a higher hit rate where the errors are small, and a method that returned the whole holohedry would also score well on the first column.

Table 9. *The true operator among the candidates, for six methods on three families of cases; every method's candidates are priced in the strain norm and kept at price* $\leq$ *band. Benchmark and stress: band* 7.4%*, the reference widened at* 8.0% *for the closure; real cells: band* 2.5 *times each cell's error strain. The last two rows give, for the closure, how often the cheapest candidate lies in the true operator's coset of the exact lattice symmetries, and the median and 90th percentile of the number of candidates. Results in* `manuscript/benchmark/results_compare.txt`*,* `results_stress.txt`*,* `results_cod.txt` *and* `results_coset.txt`*.*

| method | benchmark | stress $0.5\epsilon$ | stress $0.9\epsilon$ | real cells |
|---|---|---|---|---|
| closure | 780/780 | 60/60 | 60/60 | 210/210 |
| entries in $\{-1, 0, 1\}$ | 778/780 | 58/60 | 52/60 | 210/210 |
| entries in $\{-2, \ldots, 2\}$ | BENCHsmalltwo | 59/60 | 60/60 | 210/210 |
| cctbx similarity, range 1 | BENCHsimone | 60/60 | 60/60 | 210/210 |
| cctbx similarity, range 2 | BENCHsimtwo | 60/60 | 60/60 | 210/210 |
| lattice-symmetry pipeline | BENCHpipe | 56/60 | 54/60 | 210/210 |
| closure: true coset cheapest | 742/780 | 38/60 | 29/60 | |
| closure: candidates, median/90th percentile | BENCHcands | 4/24 | 4/18 | 2/8 |

*What the metric does not decide.* The cheapest candidate lies in the true operator's coset of the exact lattice symmetries in 742 of the 780 benchmark cases. All 20 difference-wall cases fail this test: there the cheapest candidate is the true operator composed with a pseudo-symmetry of the nearby orthorhombic corner, which costs less than the true operator at that error. In the stress cases, with errors at half the band and above, the cheapest candidate is in the true coset in 67 of 120. Price orders the candidates, and when pseudo-symmetries cost less than the error, the choice among them is left to the intensities. The guarantee is about membership, and it is membership that the closure supplies.

*Side observations.* The cases show where the Niggli cell is unstable, although they were built near zero conorms, so Niggli-specific boundaries are under-sampled. The measured cell's Niggli cell is a different cell from the reference's in 45 of 780 cases: 23 at the body-centred orthorhombic sort O1B, 12 at O3, 6 at the C-centred monoclinic sort M1B and one each at M2B, M3, R3 and T2; 14 of the 45 are a change between type I and type II. In 125 further cases the Niggli cell is the same cell in a different basis, a tie, nearly all at sorts with equal conorms (O1B 25, M1A 15, O1A 15, T1 15, M1B

13, O4 9, O2 7, R1 7, M2A 6, T2 6, O3 5, A2 1, A3 1); a tie changes the reindexing operator as surely as a different cell does. Centred references in conventional settings ($C2$, $I222$, $R3$ in hexagonal axes, $F23$, $I4_1$, $Fddd$), each measured four times in a random primitive basis with a random error of strain 0.2 to 3%, were reindexed to the conventional basis with the rational true operator among the candidates in all 24 cases.

### *5.4. Distances to the sorts*

The sort spectrum orders the possible symmetry assignments of a measured cell in one unit (Fig. 4). An oP cell $40.0 \times 40.3 \times 80\,\text{Å}$ is 1.06% from tP, the strain of its 0.75% edge mismatch. For the worked cell the nearest sorts are A2 at 0.28%, M4 (mP) at 0.78%, A3 at 3.8% and O5 (oP) at 5.1%, and every other sort is at least 25% away, a bound that holds because $\delta_1$ reaches the cap. For a tetragonal lysozyme cell indexed in $P1$ the twelve sorts from aP to tP all lie within 0.22% (supporting information, Fig. S1). Along every specialisation the distance cannot decrease, since the lower sort's subspace lies inside the upper one's, and when both projections are made in one frame, $d^2_{\text{lower}} \geq d^2_{\text{upper}} + s^2$ with $s$ the step, with equality when the non-negativity constraints are inactive. On the seven sample cells that Oishi-Tomiyasu (2024) distributes with BLDConograph, the highest Bravais type within 7.4% is the program's type in all seven, and the next type is at least 14.9% away; the program's distances in $\text{Å}^2$, divided by the mean squared edge, are within a factor of 2.3 of the strain distances (supporting information, Table S5). The type before the largest ratio between consecutive distances is the program's type in six of the seven; that ratio is a description of the spectrum, not a decision rule, and the band is the rule. COD

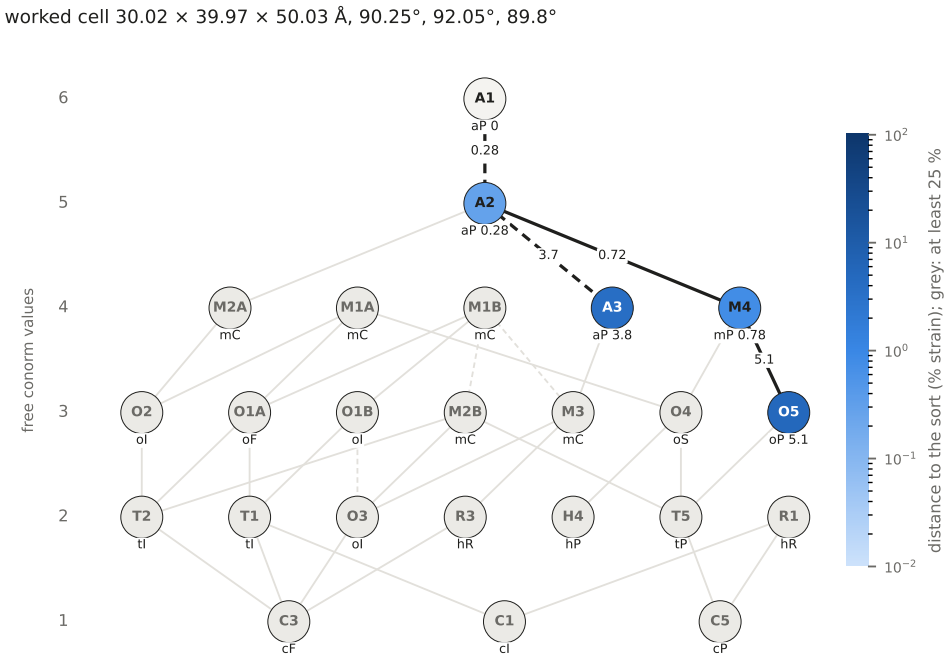



Fig. 4. The 24 Delone sorts as a graph for the measured worked cell. Rows are the number of free conorm values. Each node gives the Bravais type and the strain distance (%) from the cell to the sort, its colour the same distance on a log scale; sorts at least 25% away are grey. Edges are the specialisations of Table S2, labelled with the step from the cell idealised onto the upper sort to the lower one. Dark edges are routes on which the steps add in quadrature to the lower sort's distance within 3%, drawn where that distance is below 10%. Dashed edges keep the Bravais type.

*Against SELLA.* SELLA was run on the worked cell, the lysozyme cell, the pseudo-tetragonal cell, the 92° cell of Section 4.1, the needle, the two plates and the seven BLDConograph cells. The two rankings of the 24 sorts agree closely, with Spearman's rank correlation 0.74 to 0.97. The Bravais types within the band agree in 10 of the 14 cells, with SELLA's band set to the same relative size, 7.4% of the mean squared edge. They differ in four. On the 92° cell SELLA admits oP, at a strain of 8.85%. On the needle and the two plates an absolute band admits types that are far away in strain (for the needle hP, hR, oF, tI and tP), because one squared edge dominates the mean. On the needle and the plates $\delta_1$ is 4 to 6%, so their strain distances above that

are upper bounds and are flagged.

## 6. Discussion

*Where reindexing has been hard.* The usual pipeline reduces a cell, idealises it to the highest Bravais lattice within a tolerance and standardises it to a conventional cell. Two data sets so treated agree unless they idealise to different lattices or fall on opposite sides of a tie in the standardisation; that is where the pipeline of Table 9 lost its cases. What the symmetry step cannot propose is the flip at $ab$, because it is not a lattice symmetry. That flip separates $c$ from $a + c$, the orthogonal cell from its monoclinic-looking shapes, and the two reduced cells of the sort O3 that two measurements return. The closure contains exactly those operators, with the pseudo-symmetries, each priced. Its gain over the pipeline and over small-matrix searches is the completeness statement and the structure of the candidate set, not a higher hit rate on ordinary cells.

*The Niggli cell.* The instability of the Niggli cell (Andrews *et al.*, 1980; Andrews & Bernstein, 1988, 2014) is the same phenomenon in part. It sits at every boundary of the Niggli cone: the zero scalar products, where the cell changes between type I and type II; the ties $a = b$ and $b = c$; and the special conditions. Only the zero scalar products are zero conorms, and the closure handles them as it handles any zero conorm; the ties are equalities, and they change the basis of the Niggli cell without changing the Selling-reduced superbase. The Selling-reduced cell shares the instability at zero conorms, as Andrews *et al.* (2019*a*) note. What the closure adds is not stability of any one cell but completeness of the set of alternatives, so that two data sets reduced either way are related by a stored operator. Tolerant Niggli reduction (Křivý & Gruber, 1976; Grosse-Kunstleve *et al.*, 2004) returns one of these cells, not the set.

*Relation to $S^6$, SELLA and BLDConograph.* The flips and swaps are the boundary

transformations and reflections of $S^6$, and the closure of a lattice is the set of its images under those maps that stay in the reduced orthant (Section 2.2). The 6096 matrices of $\mathcal{U}$ are $48\times127$ on bases where $24\times127$ act on $S^6$. SELLA projects those images onto the sorts' subspaces; the sort spectrum does the same in a norm that is the same in every frame, over the frames of $\mathcal{U}$, and the two agree on most cells and differ, as Section 5.4 shows, where an absolute band treats edges of very different length alike. The norm used here is local and not a metric on lattice space; the triangle inequality between prices measured at different metrics holds with the factors of Lemma A8. NCDist and CS6Dist (Andrews & Bernstein, 2014) are metrics, which makes them suitable for searching databases; they return a distance, not the operator. Oishi-Tomiyasu (2012, 2024) determines Bravais types in Selling coordinates with a provably sufficient finite set of transforms under an error bound; the layers and adjacency radii here play the corresponding part for the reindexing operators, with the radius computed for each cell. The twofolds are the operators that matter for reindexing: their prices rank the pseudo-symmetries as the tolerance-ranked subgroups of Sauter *et al.* (2004, 2006) and the twinning operators of Lebedev *et al.* (2006) do, and the indexing classes are the twin laws of the reference, as tabulated for merohedral cases by Yeates (1997) and Dauter (2003) and resolved from intensities by Evans (2006, 2011), Brehm & Diederichs (2014) and Gildea & Winter (2018).

*Scope.* "Complete" in this paper means: every proper unimodular operator whose price is at most the band is among the candidates. The enumeration is exact for every band below 1; the layered description is exact below the adjacency radius, which is computed up to 25%. Operators that the closure does not contain are outside its scope: sublattice and superlattice relations (a doubled axis from weak odd layers; determinant $\pm2$ or more), the obverse and reverse settings of a rhombohedral lattice, which are different lattices, and the choice of hand. A user detects a doubled axis or

an obverse–reverse relation when no candidate is found at a band that should cover the error, and tests the integer sublattices of index 2 or 3 of the reference separately. A change of hexagonal basis is unimodular and is covered. An error larger than the band, which the user must estimate in advance (Section 4.5), is outside the scope too, and the choice within a class is left to the intensities.

## 7. Conclusions

All obtuse superbases of a lattice, with the integer matrices between them, form a finite groupoid: its arrows at one object are the point group, its arrows between two objects a coset of it, and no symmetry relates cells of different shape. Every arrow of every such closure lies in one set of 6096 integer matrices. For a measured cell the closure is widened to every superbase within a strain $\epsilon$ of obtuse, in a norm that is the same in every basis; the widened set is finite and is enumerated exactly. In a pipeline this means: store each reference once as its widened closure, at $\epsilon/(1-\epsilon)$ for a band $\epsilon$ set from the cells' errors and the expected non-isomorphism; Selling-reduce each new data set once and pair it with the stored members; group the proper candidates by the reference's Laue group, or its point group for anomalous data. Every proper unimodular operator of price at most $\epsilon$ is then among the candidates, each class carries a strain price, and a twofold is offered when the metrics it fixes are within $\epsilon/2$. Small-matrix and similarity searches find the true operator nearly as often on the cases tested; what the closure adds is the guarantee, the coset structure of the candidates, and the operators that are not lattice symmetries, which a symmetry search cannot propose.

**8. ***

Data availability The implementation is the module `agentsg.cell.strain_closure` of the `agentsg` repository (https://github.com/phzwart/agentsg), with the functions `widened_closure`, `reindex`, `indexing_classes`, `price_spectrum`, `sort_spectrum`, `certified_sort_spectrum` and `strain_from_esds`. Every number in this paper is reproduced by a test in `manuscript/appendix_z` or `agentsg/tests/test_strain_closure.py`; the supporting information lists them by section, table and figure (Table S8), with the versions of Python, NumPy and SciPy used and the checks that need external data. The benchmark, the comparison scripts and their outputs are in `manuscript/benchmark`.

## Appendix A
## Proofs

*Terms.* Write a lattice vector in any basis, $x = n_1 v_1 + n_2 v_2 + n_3 v_3$ with integer $n_i$, and record whether each coordinate is even or odd. This sorts the lattice into eight *parity classes*, the classes of $\Lambda/2\Lambda$; two vectors are in the same class exactly when their difference is twice a lattice vector, so the classes do not depend on the basis. One class is the doubled vectors; the other seven are used here. A *partial sum* of a superbase is the sum of some of its four vectors; there are seven up to sign, one in each of the seven classes. A superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class, that is, a Voronoi vector of the lattice (Voronoi, 1908; Conway & Sloane, 1992; Kurlin, 2022); at a zero conorm the partial sums are Voronoi vectors that are not strict. The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase $F$ in class $c$ is the squared length of its partial sum in that class minus the squared length $\mu_c$ of the shortest vector of the class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$; a vector of the same class has coordinates $s + 2k$ with $k$ integer. Every

lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{5}$$

whatever the signs of the conorms, because the four vectors sum to zero.

*Lemma A1 (window).* Let $F_0$ and $F$ be obtuse superbases of the same lattice. Then every vector of $F$ has superbase coordinates $s_i + 2k_i$ relative to $F_0$, with $s \in \{0,1\}^4$, $k_0 = 0$ and $|k_i| \le 3$. *Proof.* The non-zero conorms of $F_0$ connect its four vectors, since otherwise a partial sum would have length zero by equation (5). Every partial sum of $F$ is a shortest vector of its parity class, as is the partial sum of $F_0$ in that class, so their excess vanishes; by Lemma A2 the excess is $4\sum p^0_{ij}\gamma_{ij}$ with every term non-negative, so $\gamma_{ij} = 0$ on every non-zero slot, i.e. $|k_i - k_j| \le 1$ along a connected graph on four vertices, and with $k_0 = 0$ every $|k_i| \le 3$.□ Enumerating, over all 38 faces, the superbases of such vectors whose six pairwise products are non-positive on the whole face gives 1, 2, 3, 6 and 16 superbases for no zero, one zero, two disjoint zeros, two zeros sharing a vector and three zeros (both placements), independently of Kurlin's lists. In the basis of $F_0$ their entries are in $\{-1, 0, 1\}$ when the zero conorms avoid $d$, and reach 3 otherwise: the conorms $(0, 0, 3, 2, 0, 5)$ have the obtuse basis $(-3a-2b-c,\ a,\ a+b)$. Every member lies in $\mathcal{U}$ in each of its 48 labellings, which is the computer-assisted proof that $\mathcal{U}$ contains every arrow, and every member has a labelling with entries in $[-2, 2]$.

*Lemma A2 (class excess identity).* Let $F$ be any superbase, $w = \sum s_i v_i$ a partial sum and $w'$ a vector of its parity class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + d)$, $\Delta k = k_i - k_j$, $d = s_i - s_j$; each $\gamma_{ij}$ is a non-negative integer, zero exactly when $\Delta k \in \{0, -d\}$. *Proof.* Equation (5), $(d + 2\Delta k)^2 - d^2 = 4\Delta k(\Delta k + d)$, and $t(t + d) \ge 0$ for integers $t$ and $d \in \{-1, 0, 1\}$. □

*Lemma A3 (generation).* Every arrow of the closure, and every symmetry, is a sequence of flips at zero conorms followed by a swap. *Proof.* The obtuse superbases of

a lattice are reached from $F_0$ by flips at zero conorms (Kurlin, 2022, the lemmas listing the superbases of each Voronoi type; verified here by the enumeration of Lemma A1), so flips and swaps connect all superbases in all labellings. A symmetry maps $F_0$ to an obtuse superbase of the lattice; as a matrix it is the flip path from $F_0$ to that superbase followed by a swap. □

*Proposition A4 (the strain norm bounds every dot product).* If $\|\mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}\|_F \leq \epsilon$, then $|u^{\mathsf{T}}\Delta\mathbf{G}\,v| \leq \epsilon\,|u|\,|v|$ for all vectors $u, v$, lengths in $\mathbf{G}$; in particular every conorm of every superbase changes by at most $\epsilon|u_i||u_j|$. *Proof.* With $X = \mathbf{G}^{-1/2}\Delta\mathbf{G}\,\mathbf{G}^{-1/2}$, $u^{\mathsf{T}}\Delta\mathbf{G}\,v = (\mathbf{G}^{1/2}u)^{\mathsf{T}}X(\mathbf{G}^{1/2}v)$, at most $\|X\|_2|u||v| \leq \|X\|_F|u||v|$. □

*Definition and enumeration A5 (sorts as closed sets of equalities).* Fix a zero pattern $Z$, let $O_Z$ be the arrows of its closure and, for $\mathbf{W} \in O_Z$, let $\mathbf{A_W}$ be its linear action on conorm vectors (supporting information, Section S3), which on the face $\{p \geq 0,\ p_Z = 0\}$ is a permutation of the slots. For a subspace $V$ of the face put $\mathrm{Sym}(V) = \{\mathbf{W} \in O_Z : \mathbf{A_W} = \mathrm{id} \text{ on } V\}$ and $\mathrm{Fix}(H) = \{p : \mathbf{A_W}p = p\ \forall\,\mathbf{W} \in H\}$. A lattice with conorm vector $p$ on the face has point group $\mathrm{Sym}(\{p\})$, every symmetry being in $O_Z$ by Lemma A3. We take as the definition of a sort a maximal set of lattices with the same kind of closure and the same symmetry operators; then the sorts of kind $Z$ are the generic points of the closed subspaces $V = \mathrm{Fix}(\mathrm{Sym}(V))$ that meet the interior of the face, modulo the arrows that map the face onto itself. This is the classical construction, the fixed subspace of a point group in the space of forms (Delone *et al.*, 1975), written in conorm coordinates; SELLA uses the same subspaces. Sym and Fix are a Galois connection, so the closed subspaces are the intersections of the finitely many subspaces $\mathrm{Fix}(\mathbf{W})$; since $\mathbf{A_W}$ permutes the slots on the face, each is a set of equalities among conorms. Enumerating the intersections gives 8, 5, 3, 5 and 3 closed subspaces for no zero, one zero, two sharing a vector, two disjoint and three zeros, with symmetry orders $2, 4, 4, 8, 8, 12, 16, 48$; $2, 4, 4, 8, 16$; $4, 8, 24$; $2, 4, 8, 12, 48$; $8, 16, 48$.

Matched pattern by pattern, by computation, these are Delone's 24 sorts (Table 6).

*Lemma A6 (the widened set is finite).* If $F$ is within $\epsilon < 1$ of obtuse for $\mathbf{G}$ in the norm (2), every vector $u$ of $F$ satisfies $|u|^2_{\mathbf{G}} \leq \frac{1+\epsilon}{1-\epsilon}\,\mu_c(\mathbf{G})$, with $\mu_c$ the squared length of the shortest vector of the parity class of $u$, which is that of $F_0$'s partial sum in the class. *Proof.* Let $\mathbf{G}'$ within $\epsilon$ make $F$ obtuse. By Proposition A4 with $u = v$, $(1-\epsilon)|u|^2_{\mathbf{G}} \leq |u|^2_{\mathbf{G}'} \leq (1+\epsilon)|u|^2_{\mathbf{G}}$ for every lattice vector. The vectors of $F$ are shortest in their classes for $\mathbf{G}'$, and those of $F_0$ for $\mathbf{G}$, so $|u|^2_{\mathbf{G}'} \leq |v^0_c|^2_{\mathbf{G}'} \leq (1+\epsilon)\mu_c(\mathbf{G})$ and $|u|^2_{\mathbf{G}} \leq |u|^2_{\mathbf{G}'}/(1-\epsilon)$. □ The lattice vectors in these seven balls are finitely many and computable from $\mathbf{G}$, so $\mathcal{S}_\epsilon(\mathbf{G})$ is enumerated exactly.

*Remark A7 (layers and projection).* Two superbases are *adjacent* when some positive definite metric makes both obtuse; a superbase adjacent to $F_0$ is $\mathbf{W}F_0$ with $\mathbf{W} \in \mathcal{U}$, since both are superbases of that metric's closure. Let $\mathcal{N}_1 = \mathcal{U}F_0$, $\mathcal{N}_k = \mathcal{U}\mathcal{N}_{k-1}$, and let $\delta_k(\mathbf{G})$ be the smallest distance in the norm (2) from $\mathbf{G}$ to a metric for which a superbase outside $\mathcal{N}_k$ is obtuse. If $\epsilon < \delta_k(\mathbf{G})$, then $\mathcal{S}_\epsilon(\mathbf{G})$ is the set of members of $\mathcal{N}_k$ within $\epsilon$ of obtuse; this is the definition of $\delta_k$. The substance is the description of $\mathcal{N}_1$, which rests on the computer-assisted completeness of $\mathcal{U}$ (Lemma A1), and the projection construction: if moreover $\epsilon < \frac{1}{4}$, the projection construction of Section 4.3 returns $\mathcal{S}_\epsilon(\mathbf{G})$. *Proof of the last statement.* $F = \mathbf{W}F_0 \in \mathcal{S}_\epsilon$ is obtuse for a metric $\mathbf{G}'$ within $\epsilon$, whose zero pattern in the frame of $F$ is one of the 38 faces; the frame of $F$ is one of the frames projected onto, the projection onto that face is at distance at most $\epsilon$ from $\mathbf{G}$, and $F$ is the base superbase of its closure, so $F$ is enumerated and passes the filter. The cap $\frac{1}{4}$ keeps every projection within the margin $4\epsilon$ used in the checks non-singular, since a singular metric differs from $\mathbf{G}$ by at least 1 in the norm along its null vector. □ The radii are computed exactly from the balls of Lemma A6 at $\epsilon = \frac{1}{4}$. No lower bound on $\delta_1$ in terms of the cell's geometry is proved; the four cells of Table 7 with $\delta_1$ below the cap suggest a bound between 1 and $\sqrt{2}$ times the ratio

of the shortest to the longest edge.

*Lemma A8 (overlap: widening one side suffices).* (i) If $\|X\|_{\mathbf{A}} \le \epsilon < 1$ then $\|X\|_{\mathbf{A}+X} \le \epsilon/(1-\epsilon)$. (ii) Let $\mathbf{P}$ be an integer operator whose price measured at the new cell, $\|\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}-\mathbf{G}_N\|_{\mathbf{G}_N}$, is at most $\epsilon < 1$, and let $F$ be obtuse for $\mathbf{G}_N$. Then $\mathbf{P}F \in \mathcal{S}_{\epsilon'}(\mathbf{G}_R)$ with $\epsilon' = \epsilon/(1-\epsilon)$. *Proof.* (i) With $Y = \mathbf{A}^{-1/2}X\mathbf{A}^{-1/2}$, $\|Y\|_F \le \epsilon$, the matrix $(\mathbf{A}+X)^{-1/2}X(\mathbf{A}+X)^{-1/2}$ is orthogonally similar to $(I+Y)^{-1/2}Y(I+Y)^{-1/2}$, whose eigenvalues are $y_i/(1+y_i)$ for the eigenvalues $y_i$ of $Y$; since $|y_i| \le \epsilon$, $|y_i/(1+y_i)| \le |y_i|/(1-\epsilon)$, and the Frobenius norm is at most $\epsilon/(1-\epsilon)$. (ii) By (i) with $\mathbf{A} = \mathbf{G}_N$ and $X = \mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P} - \mathbf{G}_N$, the metric $\mathbf{G}_N$ is within $\epsilon'$ of $\mathbf{P}^{\mathsf{T}}\mathbf{G}_R\mathbf{P}$ in the norm at the latter. $F$ is obtuse for $\mathbf{G}_N$; carried back by $\mathbf{P}$, which is an isometry of the norm, $\mathbf{P}F$ is obtuse for a metric within $\epsilon'$ of $\mathbf{G}_R$ in the norm at $\mathbf{G}_R$. □ The bound in (i) is attained when $Y$ has a single non-zero eigenvalue, $-\epsilon$.

*Proposition A9 (the price of a symmetry).* Let $g$ be an integer matrix of finite order $n$, let $V_g = \{\mathbf{G} : g^{\mathsf{T}}\mathbf{G}g = \mathbf{G}\}$ be the metrics for which $g$ is a symmetry, and let $\mathbf{G}^* \in V_g$. (i) In the norm at $\mathbf{G}^*$, $\rho_{\min}\, d_{\mathbf{G}^*}(\mathbf{G}, V_g) \le \|g^{\mathsf{T}}\mathbf{G}g - \mathbf{G}\|_{\mathbf{G}^*} \le \rho_{\max}\, d_{\mathbf{G}^*}(\mathbf{G}, V_g)$, where $\rho = |e^{i\theta}-1|$ ranges over the rotation angles $\theta \ne 0$ of $X \mapsto g^{\mathsf{T}}Xg$ on symmetric matrices: $\rho = 2$ for a twofold, $\sqrt{3}$ for a threefold, $\rho \in \{\sqrt{2}, 2\}$ for a fourfold and $\rho \in \{1, \sqrt{3}\}$ for a sixfold. (ii) In the affine-invariant geodesic distance $\Delta$, the price $\Delta(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g)$ is at most $2\Delta(\mathbf{G}, V_g)$ for every $g$, and equal to it when $g$ is a twofold. *Proof.* (i) The map $\sigma : X \mapsto g^{\mathsf{T}}Xg$ on symmetric matrices has order $n$ and is an isometry of the inner product $\langle X, Y\rangle_{\mathbf{G}^*} = \mathrm{tr}(\mathbf{G}^{*-1}X\mathbf{G}^{*-1}Y)$, because $g\mathbf{G}^{*-1}g^{\mathsf{T}} = \mathbf{G}^{*-1}$. It is therefore orthogonal, the space splits orthogonally into its fixed space and planes and lines on which it rotates by $\theta = 2\pi k/n$, and $V_g - \mathbf{G}^*$ is the fixed space. Write $\mathbf{G} - \mathbf{G}^* = X_0 + X_\perp$ with $X_0$ fixed. The nearest point of $V_g$ to $\mathbf{G}$ is $\mathbf{G}^* + X_0$, at distance $\|X_\perp\|$, and $g^{\mathsf{T}}\mathbf{G}g - \mathbf{G} = (\sigma - 1)X_\perp$, whose norm is $|e^{i\theta} - 1|$ times that of each rotating component. The angles follow from the eigenvalues of $g$: those of $\sigma$

are their pairwise products. (ii) $\mathbf{G} \mapsto g^{\mathsf{T}}\mathbf{G}g$ is an isometry of $\Delta$. For any $\mathbf{G}' \in V_g$, $\Delta(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g) \le \Delta(\mathbf{G},\mathbf{G}') + \Delta(\mathbf{G}', g^{\mathsf{T}}\mathbf{G}g) = 2\Delta(\mathbf{G},\mathbf{G}')$. For a twofold, the geodesic midpoint $m$ of $\mathbf{G}$ and $g^{\mathsf{T}}\mathbf{G}g$ is unique and $g$ exchanges the two ends, so $m \in V_g$ and $\Delta(\mathbf{G}, V_g) \le \Delta(\mathbf{G}, m) = \frac{1}{2}\Delta(\mathbf{G}, g^{\mathsf{T}}\mathbf{G}g)$. □ The subspace of a sort to which $g$ belongs, read in the frame in which the sort's distance is attained, is contained in $V_{g'}$ for the conjugate $g'$ of $g$ in that frame, so the distance to the sort is at least $d(\mathbf{G}, V_{g'})$ and the price at most twice the distance to the sort; it is equal when $V_{g'}$ is that subspace, as for the unique twofold of a primitive monoclinic lattice. Prices measured at $\mathbf{G}$ rather than at $\mathbf{G}^*$ agree with (i) to first order. The common scale lies in $V_g$, so the scale fit does not change the distance to $V_g$.

*Proposition A10 (candidates and cosets).* Let $\mathbf{P}_0$ be the true operator from a reference $R$ to a new cell $N$, with price $p_0 < 1$ at $\mathbf{G}_N$; let $H_\epsilon$ be the unimodular $h$ with $\pi(h) = \|h^{\mathsf{T}}\mathbf{G}_R h - \mathbf{G}_R\|_{\mathbf{G}_R} \le \epsilon$, a finite set for $\epsilon < 1$ since $|hu|^2 \le (1+\epsilon)|u|^2$; and let $C_\epsilon$ be the candidates, the unimodular matrices of price at most $\epsilon$ at $\mathbf{G}_N$, prices taken without the scale fit. Then (i) $\pi(h_1h_2) \le \pi(h_1)+\pi(h_2)+\pi(h_1)\pi(h_2)$; (ii) $H_{(\epsilon-p_0)/(1+p_0)}\mathbf{P}_0 \subseteq C_\epsilon \subseteq H_{(\epsilon+p_0)/(1-p_0)}\mathbf{P}_0$; (iii) if no self-map of $R$ has price in $(\epsilon, 2\epsilon+\epsilon^2]$, $H_\epsilon$ is a group; (iv) if no self-map has price in $((\epsilon-p_0)/(1+p_0), (\epsilon+p_0)/(1-p_0)]$, then $C_\epsilon = H_\epsilon\mathbf{P}_0$.

*Proof.* Write $M_h = \mathbf{G}_R^{1/2} h \mathbf{G}_R^{-1/2}$, so that $\pi(h) = \|M_h^{\mathsf{T}}M_h - I\|_F$ and $\|M_h\|_2^2 \le 1+\pi(h)$. (i) $M_{h_1h_2} = M_{h_1}M_{h_2}$ and $M_{h_1h_2}^{\mathsf{T}}M_{h_1h_2} - I = M_{h_2}^{\mathsf{T}}(M_{h_1}^{\mathsf{T}}M_{h_1} - I)M_{h_2} + (M_{h_2}^{\mathsf{T}}M_{h_2} - I)$. (ii) With $\mathbf{B} = \mathbf{P}_0^{\mathsf{T}}\mathbf{G}_R\mathbf{P}_0$, $(h\mathbf{P}_0)^{\mathsf{T}}\mathbf{G}_R(h\mathbf{P}_0) - \mathbf{G}_N = \mathbf{P}_0^{\mathsf{T}}(h^{\mathsf{T}}\mathbf{G}_R h - \mathbf{G}_R)\mathbf{P}_0 + (\mathbf{B} - \mathbf{G}_N)$. The first term has norm $\pi(h)$ at $\mathbf{B}$, because $\mathbf{P}_0$ is an isometry, and its norm at $\mathbf{G}_N$ lies between $(1-p_0)\pi(h)$ and $(1+p_0)\pi(h)$, because the eigenvalues of $\mathbf{G}_N^{-1/2}\mathbf{B}\mathbf{G}_N^{-1/2}$ lie in $[1-p_0, 1+p_0]$; the second has norm $p_0$. So the price of $h\mathbf{P}_0$ lies in $[(1-p_0)\pi(h) - p_0,\ (1+p_0)\pi(h) + p_0]$, and every unimodular $\mathbf{P}$ is $h\mathbf{P}_0$ with $h = \mathbf{P}\mathbf{P}_0^{-1}$. (iii) By (i) a product of two elements of $H_\epsilon$ has price at most $2\epsilon+\epsilon^2$, hence at most $\epsilon$; a finite set of matrices closed under products is a group. (iv) Under the condition the two sets of

(ii) coincide. □ The implementation's prices include the common scale fit. For a unimodular $h$ the logarithms of the eigenvalues of $\mathbf{G}_R^{-1}h^{\mathsf{T}}\mathbf{G}_R h$ sum to zero, so the trace term that the scale removes is of second order, and the fit changes a self-map's price only at second order. The conditions in (iii) and (iv) are sufficient, not necessary: the worked cell at $\epsilon = 7.4\%$ has self-maps of price 0, 1.6% and 10%, so the condition of (iii) fails, yet $H_\epsilon = \{\mathbf{I}, 2_y\}$ is a group, which the implementation checks directly.

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**Synopsis**

Two data sets from one crystal can reduce to different cells of the same lattice, related by an integer matrix that is not a lattice symmetry. All reduced cells of a lattice and the matrices between them, widened in a basis-independent strain norm, are enumerated exactly for a measured cell and give every candidate reindexing operator within a stated tolerance, each with a price.

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